Solution:

The three year moving averages are shown in the last column.

Year	Loan by District Central Cooperative Banks	3-year moving total	3-year moving average
2004-05	41.82	-	-
2005-06	40.05	120.99	40.33
2006-07	39.12	103.89	34.63
2007-08	24.72	90.53	30.18
2008-09	26.69	111.07	37.02
2009-10	59.66	110	36.67
2010-11	23.65	111.67	37.22
2011-12	28.36	85.32	28.44
2012-13	33.31	93.27	31.09
2013-14	31.60	101.39	33.80
2014-15	36.48	-	-

Moving averages - even number of years (4 years)

1. Add up the values of the first 4 years and place the sum against the middle of 2nd and 3rd year. (This sum is called 4 year moving total)
2. Leave the first year value and add next 4 values from the 2nd year onward and write the sum against its middle position.
3. This process must be continued till the value of the last item is taken into account.
4. Add the first two 4-years moving total and write the sum against 3rd year.
5. Leave the first 4-year moving total and add the next two 4-year moving total and place it against 4th year.
6. This process must be continued till all the 4-yearly moving totals are summed up and centered.
7. Divide the 4-years moving total by 8 to get the moving averages which are our required trend values.

Example 7.6

Compute the trends by the method of moving averages, assuming that 4-year cycle is present in the following series.

Year	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008
Annual value	154.0	140.5	147.0	148.5	142.9	142.1	136.6	142.7	145.7	145.1	137.8

12th_Statistics_EM_Unit_7.indd 190 3/4/2019 9:39:22 AM